Business Intelligence – Exercise Sheet: Principal Component Analysis and Nemawashi by Hand

Bachelor in Wirtschaftsinformatik, 6th Semester

30 April 2026

Exercises on Principal Component Analysis and Nemawashi Trajectories

Exercises 1 to 15 concern manual principal component analysis. Exercises 16 to 20 concern nemawashi trajectories for organizations with three-KPI management systems.

For PCA tasks, treat each observation as a row vector and each variable as a column. Unless stated otherwise, first center every variable. Use the sample covariance matrix

$$S = \frac{1}{n-1} X_c^\top X_c.$$

Order principal components by decreasing eigenvalue. Use unit eigenvectors. The score of a centered observation x_c on loading vector u_j is

$$z_j = x_c^\top u_j.$$

If a reconstruction from the first r principal components is requested, use

$$\hat{x} = \bar{x} + \sum_{j=1}^{r} z_j u_j.$$

The explained variance ratio of principal component j is

$$\frac{\lambda_j}{\lambda_1 + \cdots + \lambda_d}.$$

For standardized PCA, first compute z-scores with the sample standard deviations and then work with the correlation matrix. The sign of an eigenvector is arbitrary. When numerical approximations are needed, round to three decimals.

For nemawashi tasks, use the normalization and plotting convention introduced in class. All KPIs in Exercises 16 to 20 are benefit-type indicators, so no inversion is required before normalization. The KPI scales are intentionally different, so do not plot the raw values directly. First normalize each KPI according to the classroom rule, then paint the trajectory in chronological order and connect successive time points with arrows.

1. **Ex 1 (20 Points):** A BI team studies weekly numbers of online orders and automatically generated invoices for three weeks.

   (a) Given
   $$A = (1, 1), \qquad B = (2, 2), \qquad C = (3, 3),$$
   compute the mean vector $\bar{x}$ and the centered observations. (4 Points)

   (b) Form the centered data matrix X_c and compute the sample covariance matrix
   $$S = \frac{1}{n-1} X_c^\top X_c.$$
   (6 Points)

   (c) Determine the eigenvalues and unit eigenvectors of S. (6 Points)

   (d) State the first principal component and explain why the second eigenvalue is zero. (4 Points)

2. **Ex 2 (20 Points):** During an ERP rollout, the team tracks manual processing effort and automation level for three process variants.

   (a) Given
   $$A = (1, 3), \qquad B = (2, 2), \qquad C = (3, 1),$$
   compute the mean vector and the centered observations. (4 Points)

   (b) Compute the sample covariance matrix S. (6 Points)

   (c) Determine the eigenvalues and unit eigenvectors of S. (6 Points)

   (d) Interpret the first principal component in business terms and explain the sign pattern of its loadings. (4 Points)

3. **Ex 3 (20 Points):** Four branches report digital-order activity with a constant system-quality score.

   (a) Given

   $$A = (1, 5), \qquad B = (2, 5), \qquad C = (4, 5), \qquad D = (5, 5),$$

   compute the mean vector and the centered observations. (4 Points)

   (b) Compute the sample covariance matrix S. (6 Points)

   (c) Determine the eigenvalues and unit eigenvectors of S. (6 Points)

   (d) Compute the one-dimensional scores of A, B, C, D on the first principal component. (4 Points)

4. **Ex 4 (20 Points):** A BI analyst has already centered two variables: app sessions and support-chat interactions. The covariance matrix is known.

   (a) Given

   $$S = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix},$$

   compute the eigenvalues of S. (6 Points)

   (b) Compute corresponding unit eigenvectors. (6 Points)

   (c) Compute the explained variance ratios of the two principal components. (4 Points)

   (d) For the centered observations

   $$p_c = (1, 1), \qquad q_c = (1, -1),$$

   compute their scores on both principal components and state which one lies entirely on PC_1 and which one lies entirely on PC_2. (4 Points)

5. **Ex 5 (20 Points):** A marketing dashboard is approximated by one principal component.

   (a) The mean vector is

   $$\bar{x} = (10, 20),$$

   and the orthonormal loading vectors are

   $$u_1 = \frac{1}{\sqrt{2}}(1, 1), \qquad u_2 = \frac{1}{\sqrt{2}}(1, -1).$$

For the centered observations

$$A_c = (-2, -2), \qquad B_c = (0, 0), \qquad C_c = (2, 2),$$

compute the scores on PC_1. (6 Points)

(b) For the centered observation

$$D_c = (3, 1),$$

compute the scores on PC_1 and PC_2. (6 Points)

(c) Reconstruct the original observation D using only PC_1. (4 Points)

(d) Compute the reconstruction error vector for D and explain what this error means geometrically. (4 Points)

6. **Ex 6 (20 Points):** A business engineer compares unstandardized and standardized PCA for monthly revenue and conversion rate.

   (a) Given

   $$A = (100, 1), \qquad B = (110, 2), \qquad C = (120, 3),$$

   compute the mean vector and the centered observations. (4 Points)

   (b) Compute the sample covariance matrix of the raw data. (6 Points)

   (c) Determine the first principal direction of the unstandardized PCA. (4 Points)

   (d) The sample standard deviations are

   $$s_1 = 10, \qquad s_2 = 1.$$

   Standardize the data, compute the correlation matrix, and determine the first principal direction after standardization. Briefly compare both results. (6 Points)

7. **Ex 7 (20 Points):** A process-mining team has already obtained the covariance matrix of two centered variables.

   (a) Given

   $$S = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix},$$

   compute the eigenvalues and unit eigenvectors. (8 Points)

(b) Compute the explained variance ratios of PC_1 and PC_2. (4 Points)

(c) For the centered observation

$$x_c = (4, 1),$$

compute the scores on both principal components. (4 Points)

(d) Reconstruct x_c using only PC_1 and compute the squared reconstruction error. (4 Points)

8. **Ex 8 (20 Points):** A retailer compares centered weekly sales and website visits in a PCA basis.

   (a) The orthonormal loading vectors are

   $$u_1 = \frac{1}{\sqrt{2}}(1, 1), \qquad u_2 = \frac{1}{\sqrt{2}}(1, -1).$$

   For

   $$\begin{aligned} A_c &= (2, 2), \qquad & B_c &= (-2, -2), \\ C_c &= (1, 1), \qquad & D_c &= (2, -2) \end{aligned}$$

   compute the scores on PC_1. (6 Points)

   (b) Compute the scores of the same observations on PC_2. (6 Points)

   (c) Identify which observation follows the common trend most strongly and which one shows the largest orthogonal deviation. (4 Points)

   (d) Reconstruct D_c using only PC_1 and explain why the one-dimensional approximation is poor. (4 Points)

9. **Ex 9 (20 Points):** A company monitors three aligned digital KPIs: orders, invoices, and deliveries.

   (a) Given

   $$A = (1, 1, 1), \qquad B = (2, 2, 2), \qquad C = (3, 3, 3),$$

   compute the mean vector and the centered observations. (4 Points)

   (b) Compute the sample covariance matrix S. (6 Points)

   (c) Determine the eigenvalues and an orthonormal set of eigenvectors. (6 Points)

(d) Explain why this three-dimensional data set can be represented exactly by one principal component. (4 Points)

10. **Ex 10 (20 Points):** A BI team has already centered three variables: revenue growth, service tickets, and self-service logins.

    (a) Given

    $$S = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix},$$

    compute the eigenvalues of S. (6 Points)

    (b) Compute an orthonormal set of eigenvectors and order them as PC_1, PC_2, PC_3. (6 Points)

    (c) Compute the explained variance ratios of the first two principal components. (4 Points)

    (d) For the centered observation

    $$x_c = (2, 1, 1),$$

    compute the scores on PC_1 and PC_2. State whether the first two principal components reconstruct x_c exactly. (4 Points)

11. **Ex 11 (20 Points):** An operations dashboard contains three centered variables with the covariance matrix below.

    (a) Given

    $$S = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 0.5 \end{bmatrix},$$

    compute the eigenvalues of S. (6 Points)

    (b) Compute corresponding unit eigenvectors. (6 Points)

    (c) Compute the cumulative explained variance ratio of the first two principal components. (4 Points)

    (d) For the centered observation

    $$x_c = (2, 0, 1),$$

    compute the scores on all three principal components and the squared reconstruction error if only the first two principal components are kept. (4 Points)

12. **Ex 12 (20 Points):** A digital-commerce dashboard uses three variables in very different units: website visits, campaign responses, and conversion rate.

    (a) Given

    $$A = (100, 10, 1), \qquad B = (200, 20, 2), \qquad C = (300, 30, 3),$$

    compute the mean vector and the centered observations. (4 Points)

    (b) Compute the sample covariance matrix of the raw data and determine the first principal direction of the unstandardized PCA. (6 Points)

    (c) The sample standard deviations are

    $$s_1 = 100, \qquad s_2 = 10, \qquad s_3 = 1.$$

    Standardize the data and compute the correlation matrix. (6 Points)

    (d) Determine the first principal direction after standardization and explain why the three variables contribute symmetrically in the standardized solution. (4 Points)

13. **Ex 13 (20 Points):** A platform team projects a three-dimensional performance vector to the first two principal components.

    (a) The mean vector is

    $$\bar{x} = (50, 50, 20),$$

    and the orthonormal loading vectors are

    $$u_1 = \frac{1}{\sqrt{2}}(1, 1, 0), \qquad u_2 = \frac{1}{\sqrt{2}}(1, -1, 0), \qquad u_3 = (0, 0, 1).$$

    For the observation

    $$x = (52, 52, 21),$$

    compute the centered observation x_c. (4 Points)

    (b) Compute the scores z_1, z_2, z_3. (6 Points)

    (c) Give the two-dimensional representation of x in the (PC_1, PC_2)-plane. (4 Points)

    (d) Reconstruct x using only PC_1 and PC_2, and compute the reconstruction error vector. (6 Points)

14. **Ex 14 (20 Points):** A BI analyst has already diagonalized a three-dimensional covariance matrix.

    (a) The eigenvalues are

    $$\lambda_1 = 5, \qquad \lambda_2 = 1.5, \qquad \lambda_3 = 0.5.$$

    Compute the explained variance ratios of the three principal components. (6 Points)

    (b) Compute the cumulative explained variance ratio after the first two principal components. (4 Points)

    (c) Determine the minimum number of principal components needed to exceed an 80% threshold and a 90% threshold. (4 Points)

    (d) The corresponding unit loading vectors are

    $$u_1 = \frac{1}{\sqrt{3}}(1,1,1), \qquad u_2 = \frac{1}{\sqrt{2}}(1,-1,0), \qquad u_3 = \frac{1}{\sqrt{6}}(1,1,-2).$$

    Interpret in words what kind of business movement each principal component captures. (6 Points)

15. **Ex 15 (20 Points):** A digital factory compares deviations in three centered indicators: automation, margin, and data quality.

    (a) Given

    $$\begin{aligned} A &= (1,0,0), \quad B = (-1,0,0), \quad C = (0,2,0), \\ D &= (0,-2,0), \quad E = (0,0,3), \quad F = (0,0,-3) \end{aligned}$$

    show that the mean vector is zero and write the centered data matrix X_c. (4 Points)

    (b) Compute the sample covariance matrix S. (6 Points)

    (c) Determine the eigenvalues and unit eigenvectors, and order them as PC_1, PC_2, PC_3. (6 Points)

    (d) Compute the explained variance ratio of the first two principal components together and decide whether a two-dimensional representation preserves at least 90% of the total variance. (4 Points)

16. **Ex 16 (20 Points):** An online retailer uses a three-KPI management system over six months $t_1, \ldots, t_6$.

    (a) The KPI series are:

K_1 Digital sales share (%): 18, 21, 25, 31, 37, 44
K_2 Average basket value (EUR): 52, 53, 52, 56, 60, 65
K_3 Monthly app sessions (thousand): 140, 155, 180, 230, 275, 330

Normalize the three KPI series according to the nemawashi classroom rule and write the normalized state for each t_i. (8 Points)

(b) Paint the nemawashi trajectory in chronological order, connect all consecutive time points with arrows, and mark the start and the end point. (8 Points)

(c) Identify one phase in which all three KPIs improve together and one phase in which one KPI lags behind the other two. (4 Points)

17. **Ex 17 (20 Points):** A smart factory program uses a three-KPI management system over seven reporting periods $t_1, \ldots, t_7$.

(a) The KPI series are:

K_1 Automated workstations (#): 8, 10, 12, 14, 17, 19, 22
K_2 First-pass yield (%): 91.0, 91.4, 92.0, 93.1, 93.8, 94.2, 95.0
K_3 Digital-maintenance savings (kEUR): 12, 13, 15, 18, 22, 27, 33

Normalize the three KPI series according to the nemawashi classroom rule and write the normalized state for each t_i. (8 Points)

(b) Paint the nemawashi trajectory, connect the time points with arrows, and label t_1 and t_7. (8 Points)

(c) Indicate whether the development looks almost linear or whether the organization changes direction visibly during the trajectory. Briefly justify your answer. (4 Points)

18. **Ex 18 (20 Points):** A B2B service platform uses a three-KPI management system over five months $t_1, \ldots, t_5$.

(a) The KPI series are:

K_1 Self-service resolution rate (%): 24, 29, 31, 34, 41
K_2 Active knowledge-base articles (#): 180, 215, 255, 300, 360
K_3 Recurring service revenue (kEUR/month): 410, 420, 438, 465, 505

Normalize the KPI series according to the nemawashi classroom rule and write the normalized state for each t_i. (8 Points)

(b) Paint the nemawashi trajectory in the correct chronological order and mark the strongest movement between two consecutive periods. (8 Points)

(c) State whether the trajectory suggests balanced improvement or whether one KPI dominates the movement. (4 Points)

19. **Ex 19 (20 Points):** A supply-chain integration program uses a three-KPI management system over eight periods $t_1, \ldots, t_8$.

    (a) The KPI series are:

    K_1 Supplier portal adoption (%): 12, 17, 23, 30, 38, 47, 55, 62
    K_2 Forecast accuracy (%): 68, 71, 74, 76, 75, 78, 81, 84
    K_3 EDI messages per week (#): 820, 950, 1120, 1290, 1480, 1710, 1980, 2290

    Normalize the KPI series according to the nemawashi classroom rule and write the normalized state for each t_i. (8 Points)

    (b) Paint the nemawashi trajectory, connect all time points with arrows, and highlight the part of the trajectory around the temporary dip in forecast accuracy. (8 Points)

    (c) Briefly interpret how the direction of the trajectory changes around that dip. (4 Points)

20. **Ex 20 (20 Points):** A sustainability and digital-product initiative uses a three-KPI management system over ten periods $t_1, \ldots, t_{10}$.

    (a) The KPI series are:

    K_1 Paperless invoice share (%): 35, 38, 41, 45, 49, 53, 58, 63, 68, 72
    K_2 CO_2 data records per month (#): 1200, 1320, 1480, 1690, 1940, 2230, 2580, 3000, 3470, 3970
    K_3 Green product revenue (kEUR): 180, 188, 200, 214, 231, 252, 278, 309, 345, 388

    Normalize the KPI series according to the nemawashi classroom rule and write the normalized state for each t_i. (8 Points)

    (b) Paint the nemawashi trajectory in chronological order, mark the start and end state, and indicate any phase in which the trajectory becomes noticeably steeper. (8 Points)

    (c) State whether the trajectory appears smooth over time or whether it contains a structural jump. Briefly justify your answer. (4 Points)